



7321 - SN 2008S and NGC 300 2008-OT: Are They Terminal Electron Capture Supernovae?

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
SN2008S				
NGC300-2008OT				
	3		MIRI Imaging	(2) NGC-300-OT2008-1
	4		NIRCam Imaging	(2) NGC-300-OT2008-1

ABSTRACT

SN 2008S and NGC 300 2008-OT are the archetypical members of a class of highly reddened transients called intermediate luminosity red transients (ILRTs). All observed ILRT progenitors, including the progenitors of SN 2008S and NGC 300 2008-OT, are heavily obscured by circumstellar dust and the transients themselves are also very dusty. It is unknown what physical mechanisms produce ILRTs. One proposed explanation for ILRTs is that they are electron capture supernovae (ECSNe) resulting from the core-collapse of eight to ten solar mass super asymptotic branch stars (SAGB). We propose to obtain very late time infrared photometry of SN 2008S and NGC 300 2008-OT to investigate whether or not ILRTs have a luminous remnant. With the sensitivity of JWST in the infrared we are able to detect dust-obscured luminous remnants down to 10% the luminosity of their

progenitors. No detection would confirm that ILRTs are terminal ECSNe.

OBSERVING DESCRIPTION

We are proposing to obtain very late-time photometry for SN 2008S and NGC 300 2008-OT (the two archetypical intermediate luminosity red transients) with MIRI and NIRCам in 8 filters ranging in wavelength from 1-21 microns. In total, including overheads, we are requesting 11.9 hours of JWST time.

Proposal 7321 - Targets - SN 2008S and NGC 300 2008-OT: Are They Terminal Electron Capture Supernovae?

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(2)	NGC-300-OT2008-1	RA: 00 54 34.1600 (13.6423333d) Dec: -37 38 28.60 (-37.64128d) Equinox: J2000	Epoch of Position: 2000	
	<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>Category=Star</i> <i>Description=[Circumstellar dust, Cocoon stars, Supernovae]</i>				

Proposal 7321 - Observation 3 - SN 2008S and NGC 300 2008-OT: Are They Terminal Electron Capture Supernovae?

Thu Sep 04 22:00:08 GMT 2025

Observation	Proposal 7321, Observation 3									
	Diagnostic Status: Warning Observing Template: MIRI Imaging									
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous	
	(2)	NGC-300-OT2008-1	RA: 00 54 34.1600 (13.6423333d) Dec: -37 38 28.60 (-37.64128d) Equinox: J2000			Epoch of Position: 2000				
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Circumstellar dust, Cocoon stars, Supernovae]									
Template	Subarray									
	FULL									
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size
	1	CYCLING	1	4						DEFAULT
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time
	1	F770W	FASTR1	100	2	1	Dither 1	4	8	2231.132
	2	F1000W	FASTR1	100	2	1	Dither 1	4	8	2231.132
	3	F1500W	FASTR1	100	2	1	Dither 1	4	8	2231.132
	4	F2100W	FASTR1	25	8	1	Dither 1	4	32	2297.733
Special Requirements	Group Observations 3, 4, Non-interruptible									

Proposal 7321 - Observation 4 - SN 2008S and NGC 300 2008-OT: Are They Terminal Electron Capture Supernovae?

Observation	Proposal 7321, Observation 4										Thu Sep 04 22:00:08 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	NGC-300-OT2008-1	RA: 00 54 34.1600 (13.6423333d)			Epoch of Position: 2000					
			Dec: -37 38 28.60 (-37.64128d)								
			Equinox: J2000								
			Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.								
Template		Category=Star									
		Description=[Circumstellar dust, Cocoon stars, Supernovae]									
Template	Module					Subarray					
	B					FULL					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRASCA		4		STANDARD		8" (SMALL)		1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F115W	F356W	SHALLOW4	8	1	4	4	1674.936		
	2	F200W	F444W	SHALLOW4	8	1	4	4	1674.936		
Special Requirements	Offset 38.0 arcsec, 38.0 arcsec										
	Group Observations 3, 4, Non-interruptible										